Green-function Zeros Encode Competing Mott and Charge-ordering Scales

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While correlated insulators are devoid of low-energy quasiparticle poles, their Green functions retain clean momentum structure through zeros. However, precisely what zeros imply is not clear. By studying the extended Hubbard model both analytically and numerically, we establish a new paradigm for strongly correlated matter: the dispersion of Green function zeros is determined by both microscopic spin-spin correlations and defect kinematics. In fact, we find that the dispersion changes discontinuously across the transition between the Mott and the checkerboard charge-density wave phases as is expected for a first-order transition. This physics is robust to the inclusion of further neighbor hopping which simply fine tunes the spin-correlation or charge-defect kinematics. We conclude that it is the *dispersion* of the Green-function zeros that encode the physics of ordering resultant from the strong correlations.

INTRODUCTION

Since their inception, Mott[1] insulators have always been associated with ordering. This is natural because a half-filled band with spin-1/2 electrons in the square lattice will order antiferromagnetically at sufficiently low temperature. However, as Anderson[2] has emphasized, the ordering that ensues is a consequence of the strong correlations not the root cause of the insulating state. Precisely how to disentangle ordering from the insulating state has been the persistent question of Mott insulation. In 2003, Dzyaloshinskii[3] pointed out that zeros, of the single-particle Green function can obtain in a Mott insulator. In essence, zeros arise from a balancing act between spectral weight above and below the chemical potential. While poles count quasiparticles, it is unclear what zeros entail. Nonetheless, because cuprate superconductors start off as Mott insulators, the precise meaning of zeros is a hotly debated topic[4–15]. A point of concurrence among the different approaches is that zeros emerge from a divergence of the self energy and hence lie outside perturbation theory. Such breakdowns can lead to non-analyticities in the Luttinger-Ward functional which can generate a triangle-diagram anomaly[16, 17] in the Luttinger count.

In general, the zero-crossings will depend on momentum and frequency. Hence, what is needed to delineate the full zeros spectrum (not just the zero-frequency crossing which defines the Luttinger-Ward[18] surface) is a sufficiently non-perturbative method that has momentum and real-frequency resolution. Consequently, methods that require analytic continuation and are susceptible to the sign problem are difficult to resolve zero crossings. A few of the methods employed include numerics[19, 20] and diagrammatic expansions[21] which yield a dispersion of the form ϵ_{k+Q}/U^2 where ϵ_{k+Q} is the bare band energy shifted by the M-point momentum and U is the on-site repulsion. A distinct feature of this dispersion is that it vanishes in the $U = \infty$ limit. That is, the zeros do not disperse when the correlations are infinitely strong, reminiscent of the dynamical mean-field theory (DMFT)[22] result of dispersionless zeros at $\omega = 0$. A study of the Su-Schrieffer-Heeger-Hubbard[23] model finds spin-spin correlations play a role in determining the magnitude of the zero dispersion, whereas the band Hatsugai-Kohmoto[24, 25] (HK) model yields a dispersing band of zeros determined solely by the non-ineracting $\varepsilon(\mathbf{k})$.

To arrive at a general conclusion regarding zeros, we focus on a model that has two competing ordering tendencies, Mott antiferromagnetism and charge-density-wave (CDW) ordering shown in Fig. (1). Hence our conclusions apply beyond Mott physics. Our tool is a generalization of the recently developed[26] momentum-mixing Hatsugai-Kohmoto (MMHK) framework to include nearest-neighbor (NN) repulsion V, enabling a dynamical study of the competition between spin-correlated Mott and checkerboard CDW states. As demonstrated[26] previously, this framework interpolates systematically between the exactly solvable HK limit [24] and the Hubbard model while preserving the HK fixed-point structure [27]. What we explore here is the here-to-fore unearthed real-frequency spectral evolution.

The key result we derive is a dispersion (not just the zero-frequency crossing discussed by Dzyaloshinskii[3]) relationship for the gap-interior zeros across the Mott–CDW transition. Quite generally, we show that the simple relationship $\omega_{\rm GZ}(\mathbf{k}) \simeq s_{\rm NN}\varepsilon_{\rm NN}(\mathbf{k})$ captures the dispersion of the gap-interior zeros. Here $s_{\rm NN}$ is a numerical factor determined by the strong-correlation spin or charge ordering physics and $\varepsilon_{\rm NN}(\mathbf{k})$ is the non-interacting band dispersion. Because on the Mott side, $s_{\rm NN} \approx -1.4$, whereas on the CDW side, $s_{\rm NN} = -8/7$, we find that the bandwidth of the zeros remains finite in the infinite-interaction limit in contrast to the diagrammatic extension of DMFT[21] results discussed earlier. Local spin correlations are absent from band HK, resulting in

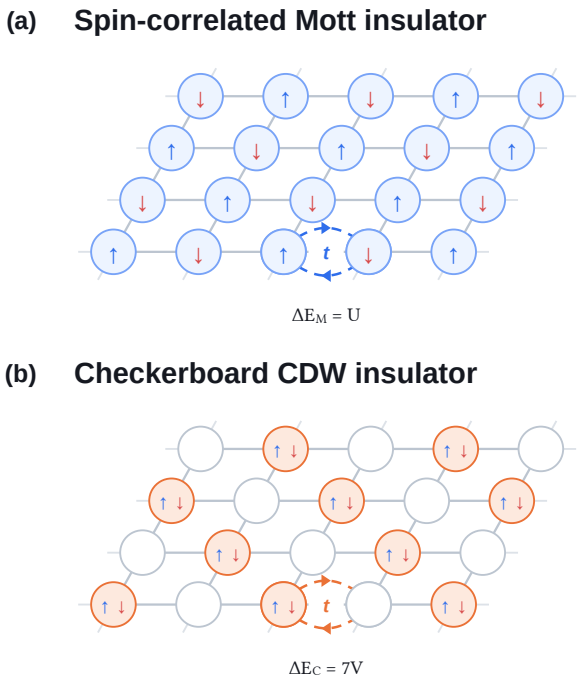


FIG. 1. **Distinct correlated insulators.** (a) In the spin-correlated Mott insulator, sites are predominantly singly occupied, and NN hopping t virtually accesses a doublon–holon configuration at energy cost $\Delta E_{\mathrm{M}} = U$ with $V = 0$. (b) In the checkerboard CDW insulator, doubly occupied and empty sites alternate, and the corresponding virtual hop accesses two singly occupied defects at cost $\Delta E_{\mathrm{C}} = 7V$ with $U = 0$. Dashed counterpropagating arrows identify the virtual process on a representative bond; the finite patches illustrate the local backgrounds.

$s_{\mathrm{NN}} = 1$. The fact that in both MMHK and HK the width of the zero dispersion remains finite points to the underlying Z_2 universality[28]. The value of s_{NN} on both phases is verified numerically and analytically. The discontinuous jump in s_{NN} and the independent discontinuities in the charge and spin correlation establish the first-order nature of the Mott–CDW transition. Hence, we arrive at the surprising result that the GZ dispersion is determined by the underlying order that accompanies Mott–CDW physics. Consequently, it is dispersive zeros that complete the Mott story and distinguish it from perturbative Slater physics.

V-EXTENDED MMHK MODEL AND GZ DISPERSION

We first define the V-extended MMHK construction at a general momentum-mixing resolution. We define $\mathcal{C}_{n_{\mathrm{mix}}} = \{\mathbf{r}_\alpha\}_{\alpha=1}^{n_{\mathrm{mix}}}$ as an n_{mix}-orbital mixing cell. In reciprocal space, translations of the reduced Brillouin zone ($\mathrm{rBZ}_{n_{\mathrm{mix}}}$) by the n_{mix} discrete mixing momenta $\mathbf{K}$ tile the full Brillouin zone. The continuous vector $\mathbf{f} \in \mathrm{rBZ}_{n_{\mathrm{mix}}}$ selects the same relative point within every tile, so a fixed-$\mathbf{f}$ sector contains the n_{mix} physical momenta $\mathbf{k} = \mathbf{K} + \mathbf{f}$. The operator $c_{\mathbf{f}\alpha\sigma}$ annihilates a spin-σ fermion in the orbital α with momentum $\mathbf{f}$. The V-extended n_{mix}-MMHK Hamiltonian is

$$H^{(n_{\mathrm{mix}})}_{\mathrm{MMHK}} = \sum_{\mathbf{f}\in\mathrm{rBZ}_{n_{\mathrm{mix}}}} \Bigg[\sum_{\alpha,\beta,\sigma} g^{t,t'}_{\alpha\beta}(\mathbf{f})\, c^\dagger_{\mathbf{f}\alpha\sigma} c_{\mathbf{f}\beta\sigma} - \mu \sum_{\alpha\sigma} n_{\mathbf{f}\alpha\sigma} + U \sum_{\alpha} n_{\mathbf{f}\alpha\uparrow} n_{\mathbf{f}\alpha\downarrow} + V \sum_{\langle\alpha\beta\rangle} n_{\mathbf{f}\alpha} n_{\mathbf{f}\beta} \Bigg]. \tag{1}$$

Here $g^{t,t'}_{\alpha\beta}(\mathbf{f}) = \sum_{\Delta\mathbf{R}} t_{\alpha\beta}(\Delta\mathbf{R})\, e^{i\mathbf{f}\cdot(\mathbf{r}_\alpha - \mathbf{r}_\beta + \Delta\mathbf{R})}$ and $t_{\alpha\beta}(\Delta\mathbf{R})$ contains the NN and next-nearest-neighbor (NNN) hopping. The bond sum $\langle\alpha\beta\rangle$ contains each NN bond of the periodic internal mixing cell once. Crucially, the interaction operators in Eq. (1) carry the same sector momentum $\mathbf{f}$ as the kinetic operators; suppressing that label would obscure the defining MMHK decomposition. For $V = 0$ and $n_{\mathrm{mix}} = 1$, Eq. (1) reduces to the solvable HK limit. Increasing n_{mix} restores progressively more interaction-induced momentum-transfer processes. At fixed lattice size N, $n_{\mathrm{mix}} = N$ makes the reduced zone a single point and exactly recovers the Hubbard interaction; the same completeness argument applied to the NN form factor $\mathcal{V}(\mathbf{q}) = 2V(\cos q_x + \cos q_y)$ recovers $V\sum_{\langle ij\rangle} n_i n_j$. The full V-extended construction and its extended-Hubbard limit are derived in Supplemental Material, Sec. 1 [26].

This manuscript uses $n_{\mathrm{mix}} = 16$ with a 4×4 mixing cell, $\alpha \equiv \mathbf{r} = (x, y)$ and $x, y = 0, 1, 2, 3$. The corresponding scattering momenta are $\mathcal{K}_{16} = \{(\pi/2)(m, n) : m, n = 0, 1, 2, 3\}$. For a continuous sector offset $\mathbf{f}$, the physical momentum is $\mathbf{k} = \mathbf{K} + \mathbf{f}$ ($\mathbf{K} \in \mathcal{K}_{16}$ and $\mathbf{f} \in \mathrm{rBZ}_{16}$). After Fourier transforming the internal coordinate, we see that the hopping term in Eq. (1) yields a bare dispersion $\varepsilon(\mathbf{k}) = \varepsilon_{\mathrm{NN}}(\mathbf{k}) + \varepsilon_{\mathrm{NNN}}(\mathbf{k})$, where $\varepsilon_{\mathrm{NN}}(\mathbf{k}) = -2t(\cos k_x + \cos k_y)$ and $\varepsilon_{\mathrm{NNN}}(\mathbf{k}) = -4t' \cos k_x \cos k_y$ are the NN and NNN hopping contributions, respectively. Continuous $\mathbf{f}$ supplies thermodynamic momentum resolution, whereas 4×4 specifies the interaction-mixing resolution. We work at half filling with $\mu = U/2 + 4V$. We first set $t' = 0$ to establish the simple GZ-dispersion relation under strong correlations in the particle–hole-symmetric case. We then turn on t' to generalize it to a generic band without particle–hole symmetry, for which the GZ dispersion separates into distinct hopping contributions.

The central observable is the continuous low-spectral-weight solution to

$$\mathrm{Re}\, G[\mathbf{k}, \omega_{\mathrm{GZ}}(\mathbf{k})] = 0,\ \omega_{\mathrm{GZ}}(\mathbf{k}) \text{ inside the central gap.} \tag{2}$$

$\omega_{\mathrm{GZ}}(\mathbf{k})$ denotes the **GZ dispersion**. Finite numerical broadening leaves a small residual $\mathrm{Im}\, G$ inside the

gap that vanishes as the broadening is removed. In the zero-broadening limit, the $\omega = 0$ intersection of this dispersion—the set of momenta satisfying $\omega_{\mathrm{GZ}}(\mathbf{k}) = 0$—is the conventional Luttinger surface $G(\mathbf{k}, 0) = 0$. A previous study[5] demonstrated that the Luttinger surface is pinned at the Fermi surface of the non-interacting system in the presence of particle-hole symmetry. It is the general behaviour of ω_{GZ} which is not necessarily pinned to the Luttinger surface that we calculate here.

RESULTS

Benchmark validation

We first benchmark 16-MMHK against state-of-the-art Hubbard-model results for the ground-state energy and double occupancy at half filling and two doped densities in Fig. 2(a,b). The reference calculations use AFQMC and DMET [29–31]. The mean deviation in the ground-state energy across the displayed interaction values is below 2% at each density. The corresponding double-occupancy deviation is below 2% at half filling and below 5% for the doped densities.

We next test the more demanding long-range magnetic response in Fig. 2(c), where $m = [S^{zz}(\mathbf{Q} = (\pi, \pi))]^{1/2}$ at half filling. Its quantitative spread relative to AFQMC and DMET is larger than that for the energy and double occupancy, but the 16-MMHK sequence increases monotonically with U and approaches the Heisenberg limit expected as $U/t \to \infty$ [32, 33]. In contrast, the displayed AFQMC sequence is non-monotonic and the DMET value at $U/t = 8$ lies above that limit, highlighting magnetic order as a more sensitive test of 16-MMHK. Independent spinless t–V tests of the added nearest-neighbor interaction—mixing-resolution convergence in one dimension and comparison with continuous-time-QMC and DMRG in two dimensions—are presented in Supplemental Material, Sec. 2 and Fig. S1. Together with the accurate ground-state energy and double occupancy and the physically consistent magnetic trend above, these tests establish the static baseline for applying V-extended 16-MMHK to the real-frequency GZ problem.

Spin and charge backgrounds predict distinct GZ coefficients

With the static benchmarks in hand, we turn to the half-filled real-frequency Green function and its zeros computed with the V-extended 16-MMHK construction. All spectral maps use the fixed broadening $\eta = 0.05$. The U-only data span $U = 8, 12, 20$, and 40 with $V = 0$; representative spectra at $U = 8$ and 20 are shown in Fig. 3(a,c), while panel (b) compares the GZ curves for all

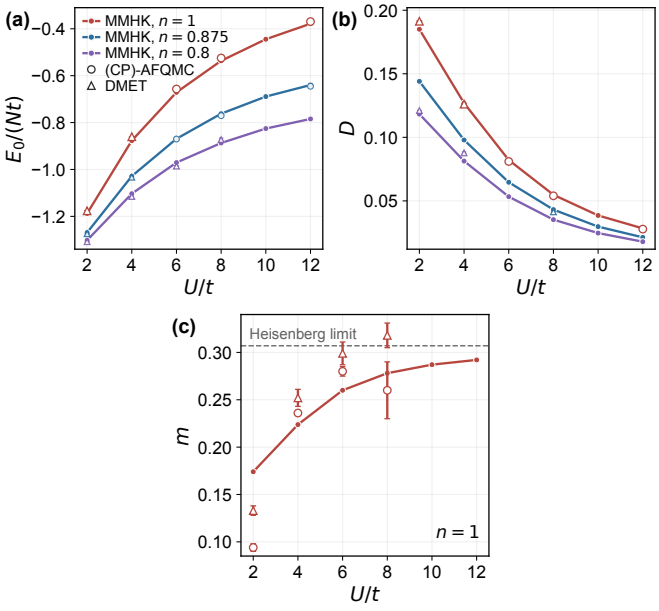


FIG. 2. **Benchmark validation of 16-MMHK.** (a, b) Ground-state energy and double occupancy for $n = 1$, 0.875, and 0.8. Color identifies density; filled circles connected by lines are MMHK, open circles are AFQMC or constrained-path AFQMC, and open triangles are DMET. (c) Half-filled staggered magnetization; the Heisenberg limit is shown. Benchmark-source selection, pointwise values, deviations, and source identities are reported in the Supplemental Material, Sec. 2 and Table S1.

four interactions. The V-only data span $V = 2, 5, 8$, and 12 at $U = 0$; representative spectra at $V = 2$ and 5 appear in Fig. 4(a,c), with all four GZ curves in panel (b). The white lines mark smooth, single-valued gap-interior GZ dispersions despite the broad Hubbard continua on the Mott side and the narrowing charge-defect bands on the CDW side. We first establish why a zero must obtain and derive what fixes its leading dispersion.

In either correlated insulator, the interaction separates the single-particle spectrum into removal and addition sectors. For each resolved momentum in the present Lehmann representation, $\mathrm{Re}\, G$ is positive immediately above the highest removal pole and negative immediately below the lowest addition pole; continuity therefore enforces at least one zero in the intervening gap. Because $G^{-1} = G_0^{-1} - \Sigma$, a GZ corresponds to a self-energy pole, a nonperturbative signature of strong correlation.

At $t' = 0$, the numerically extracted GZ dispersions reveal a common simplification. As shown for the U-only and V-only sequences in Fig. 3(b) and Fig. 4(b), respectively, their momentum dependence is accurately described by,

$$\omega_{\mathrm{GZ}}(\mathbf{k}) = s_{\mathrm{NN}} \varepsilon_{\mathrm{NN}}(\mathbf{k}), \tag{3}$$

where $s_{\mathrm{NN}}(U, V)$ is the signed, dimensionless, and $\mathbf{k}$-independent NN coefficient. Consequently, the zero-

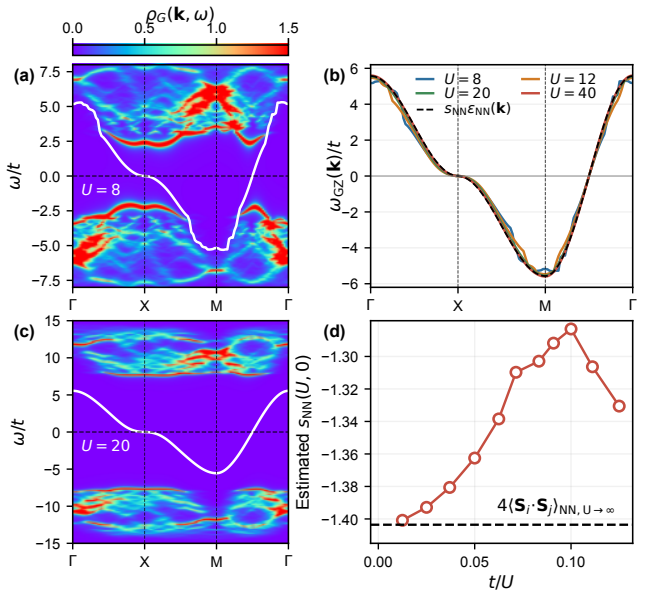


FIG. 3. **Mott-side GZ dispersion.** (a,c) Representative spectral functions at $V = 0$ for $U = 8$ and 20; white curves mark the gap-interior GZ dispersion. The frequency range expands with the interaction while the spectral color scale is fixed. (b) Extracted GZ dispersions for $U = 8, 20$, and 40 and the zero-offset fit $\omega_{GZ}(\mathbf{k}) = s_{NN}\varepsilon_{NN}(\mathbf{k})$ to the $U = 40$ curve. (d) Estimated $s_{NN}(U, 0)$ versus t/U for $U = 8 \sim 80$. The dashed line is the independently determined prediction $4\langle \mathbf{S}_i \cdot \mathbf{S}_j \rangle_{NN,U\to\infty} = -1.4036$ at the present 16-MMHK resolution. The complete spectral sequence is shown in Supplemental Fig. S2.

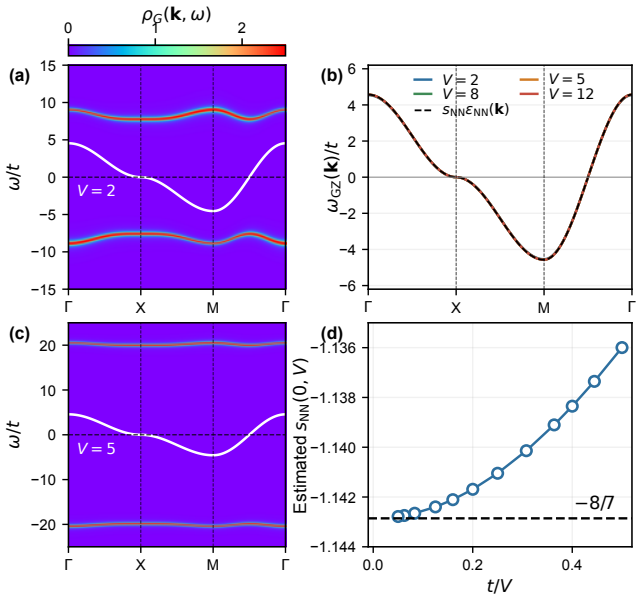


FIG. 4. **CDW-side GZ dispersion.** (a,c) Representative spectral functions at $U = 0$ for $V = 2$ and 5; white curves mark the gap-interior GZ dispersion. The frequency range expands with the interaction while the spectral color scale is fixed. (b) Extracted GZ dispersions for $V = 2, 5$, and 8 and the zero-offset fit $\omega_{GZ}(\mathbf{k}) = s_{NN}\varepsilon_{NN}(\mathbf{k})$ to the $V = 8$ curve. (d) Estimated $s_{NN}(0, V)$ versus t/V for $V = 2 \sim 20$. The dashed line is the parameter-free charge-defect prediction $-8/7$. The complete spectral sequence is shown in Supplemental Fig. S3.

frequency contour $\omega_{GZ}(\mathbf{k}) = 0$ coincides with $\varepsilon_{NN}(\mathbf{k}) = 0$, so the Luttinger surface is identical to the noninteracting Fermi surface at half filling, consistent with earlier work [5]. The value and physical origin of s_{NN} can be determined analytically from the first spectral moments of the particle-removal and particle-addition sectors:

$$G(\mathbf{k},\omega) \simeq \frac{n_{\mathbf{k}\sigma}}{\omega - E_h(\mathbf{k})} + \frac{1 - n_{\mathbf{k}\sigma}}{\omega - E_p(\mathbf{k})}. \tag{4}$$

Here E_h and E_p are the centroids of the complete many-pole removal and addition sectors. The zeroth-moment sum rules identify their integrated weights as $n_{\mathbf{k}\sigma}$ and $1 - n_{\mathbf{k}\sigma}$, respectively, where $n_{\mathbf{k}\sigma} = \langle c^\dagger_{\mathbf{k}\sigma} c_{\mathbf{k}\sigma} \rangle$ is the spin-resolved momentum distribution. We do not claim that the spectrum contains only two poles; we use the sector-centroid expansion only to locate the leading gap-interior GZ. Setting Eq. (4) to zero gives

$$\omega_{GZ} \simeq n_{\mathbf{k}\sigma} E_p + (1 - n_{\mathbf{k}\sigma}) E_h = \mathcal{C}_{\mathbf{k}} + \delta_{\mathbf{k}} \Delta_{\mathbf{k}}, \tag{5}$$

where $\mathcal{C}_{\mathbf{k}} = (E_p + E_h)/2$ is the midpoint of the sector centroids relative to the chemical potential, $\delta_{\mathbf{k}} = n_{\mathbf{k}\sigma} - 1/2$ is the deviation of the spin-resolved momentum distribution from half occupancy, and $\Delta_{\mathbf{k}} = E_p - E_h$ is the centroid separation. Equation (5) therefore separates the GZ energy into the common sector shift and the shift produced by unequal weights across the interaction-scale separation. In the atomic two-pole problem, this relation is exact and coincides with the pole in the self-energy of Ref. [7]; here it provides the leading term of a strong-coupling spectral-moment expansion [34, 35]. Because $\Delta_{\mathbf{k}}$ is $O(U)$ or $O(V)$ while the virtual-hopping weight imbalance $\delta_{\mathbf{k}}$ is $O(t/U)$ or $O(t/V)$, their product in Eq. (5) remains $O(t)$. The full moment derivation and corrections are given in Supplemental Material, Sec. 4.

For $U \gg t$ and $V = 0$, the spin-correlated Mott background in Fig. 1(a) is predominantly singly occupied, so hopping between neighboring sites creates a virtual doublon–holon pair governed by the bond spin correlation $C_{NN} = \langle \mathbf{S}_i \cdot \mathbf{S}_{i+\hat{x}} \rangle$. The Schrieffer–Wolff t/U expansion [36] gives $\mathcal{C}_{\mathbf{k}} = (1/2 + 2C_{NN})\varepsilon_{NN}(\mathbf{k})$ and $\delta_{\mathbf{k}}\Delta_{\mathbf{k}} = (2C_{NN} - 1/2)\varepsilon_{NN}(\mathbf{k})$. The spin-independent $\pm\varepsilon_{NN}(\mathbf{k})/2$ terms cancel, whereas the $2C_{NN}\varepsilon_{NN}(\mathbf{k})$ terms add, yielding

$$\omega^{(U)}_{GZ}(\mathbf{k}) = 4C_{NN}\varepsilon_{NN}(\mathbf{k}) + O(t^2/U). \tag{6}$$

Since $C_{NN} < 0$, Eq. (6) predicts the signed coefficient $s_{NN} = 4C_{NN} < 0$ without fitting the GZ curve. Details are given in Supplemental Material, Sec. 5. Figure 3 tests Eq. (6) against the full Green function. Pan-

els (a,c) show broad Hubbard continua coexisting with a clean gap-interior GZ, while panels (b,d) show its NN momentum dependence and coefficient approaching $4C_{\mathrm{NN}}$. The relation persists down to $U = 8t$ and is therefore not confined to the formal $U \to \infty$ limit. Additional spectral snapshots and occupied-band views are provided in Supplemental Material, Sec. 3 and Figs. S2 and S4.

For $V \gg t$ and $U = 0$, Fig. 1(b) shows the checkerboard background of alternating doubly occupied and empty sites. A hop across a doublon–holon bond creates a defect costing $3(2V)+V = 7V$, and therefore produces the weight imbalance $\delta_{\mathbf{k}} = n_{\mathbf{k}\sigma} - 1/2 = -\varepsilon_{\mathrm{NN}}(\mathbf{k})/(7V)$. There is no accompanying spin-correlation factor: on a doublon–empty bond, the allowed-hop projector is exactly $n_{i\sigma}(1 - n_{j\sigma}) = 1$, and both atomic sites carry zero spin. The particle-addition and particle-removal moments are separated by $8V$.

Defining the signed first moment as $\mathcal{M}_1 = (1 - n_{\mathbf{k}\sigma})E_p + n_{\mathbf{k}\sigma}E_h$ (the complementary weighting to Eq. (5)), the exact extended-Hubbard commutator gives, to leading order in t/V, $\mathcal{M}_1(\mathbf{k}) = \varepsilon_{\mathrm{NN}}(\mathbf{k}) - V\delta_{\mathbf{k}} + O(t^2/V)$. Together with $\mathcal{M}_1 = \mathcal{C}_{\mathbf{k}} - \delta_{\mathbf{k}}\Delta_{\mathbf{k}}$, this gives $\mathcal{C}_{\mathbf{k}} = \varepsilon_{\mathrm{NN}}(\mathbf{k}) + 7V\delta_{\mathbf{k}} = O(t^2/V)$. Unlike the Mott case, here the sector center is $\mathcal{C}_{\mathbf{k}}$ dispersionless. The GZ nevertheless remains dispersive at $O(t)$, because its leading contribution comes entirely from the spectral-weight term $\delta_{\mathbf{k}}\Delta_{\mathbf{k}}$. Equation 5 then gives

$$\omega_{\mathrm{GZ}}^{(V)}(\mathbf{k}) = -\frac{8}{7}\varepsilon_{\mathrm{NN}}(\mathbf{k}) + O(t^2/V). \tag{7}$$

Equation (7) fixes the parameter-free coefficient $s_{\mathrm{NN}}(0, V \to \infty) = -8/7$ from local charge-defect geometry. The full derivation is given in Supplemental Material, Sec. 6.

Figure 4 tests this prediction numerically. Panels (a,c) show the particle-addition and particle-removal features narrowing toward charge-defect bands while the gap-interior GZ remains smooth; panels (b,d) show its NN momentum dependence and coefficient converging to $-8/7$. Together with Fig. 3, these results establish two distinct strong-coupling mechanisms within the same sector-cancellation structure of Eq. (5): spin exchange on the Mott side and local defect kinematics on the CDW side. Additional spectra and occupied-band views are provided in Supplemental Figs. S3 and S5.

Finite-(U, V) strong-coupling laws

Figure 5(a) shows that $|s_{\mathrm{NN}}|$ grows on the Mott side and switches discontinuously to the CDW branch as V increases, although the central spectral gap stays open (Supplemental Material, Sec. 7 and Figs. S6 and S7). This jump reflects a change in the virtual charge defect that controls the GZ dispersion. Combining the

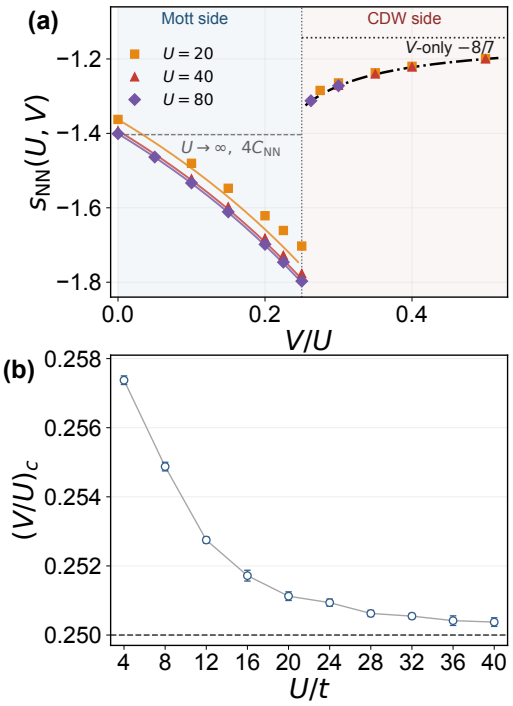


FIG. 5. **GZ coefficient s_{NN} across the Mott–CDW boundary.** (a) Signed coefficients for $U/t = 20$, 40, and 80. Symbols are the GZ data. On the Mott side, the colored curves use Eq. (8) with the independently calculated $C_{\mathrm{NN}}(U, V)$; the black dash–dotted curve on the CDW side uses the leading rational term of Eq. (9). Horizontal guides mark the interaction-axis limits and the vertical dotted line marks $V/U = 1/4$. (b) Independently bracketed first-order transition ratio. Bars show the sampled brackets, the gray line guides the eye, and the dashed line marks 1/4; Supplemental Material gives the ground-state diagnostics.

exact spectral first moment with the leading NN hopping coherence determines the strong-coupling law on each side. The notation distinguishes the neutral virtual-defect costs ΔE_{M} and ΔE_{C}, which control the hopping coherence, from the addition–removal sector separations Δ_{M} and Δ_{C} entering the moment relation. On the Mott side, the virtual NN doublon–holon excitation has energy cost $\Delta E_{\mathrm{M}} = U - V$. The corresponding spectral-weight imbalance is $\delta_{\mathbf{k}} = (2C_{\mathrm{NN}} - 1/2)\varepsilon_{\mathrm{NN}}(\mathbf{k})/\Delta E_{\mathrm{M}}$, while the particle–hole separation is $\Delta_{\mathrm{M}} = U$. To leading order, the exact first moment becomes $\mathcal{M}_1 = \varepsilon_{\mathrm{NN}}(\mathbf{k}) - V\delta_{\mathbf{k}} + O(t^2/\Delta E_{\mathrm{M}})$; substituting it into $\omega_{\mathrm{GZ}} = \mathcal{M}_1 + 2\delta_{\mathbf{k}}\Delta_{\mathrm{M}}$ gives

$$s_{\mathrm{NN}}^{\mathrm{M}}(U, V) = 1 + \frac{2U - V}{U - V}\left(2C_{\mathrm{NN}} - \frac{1}{2}\right) + O\left(\frac{t}{\Delta E_{\mathrm{M}}}\right). \tag{8}$$

The Mott-side curves in Figure 5(a) are obtained by evaluating Eq. (8) with the independently calculated $C_{\mathrm{NN}}(U, V)$.

On the CDW side, the relevant checkerboard defect

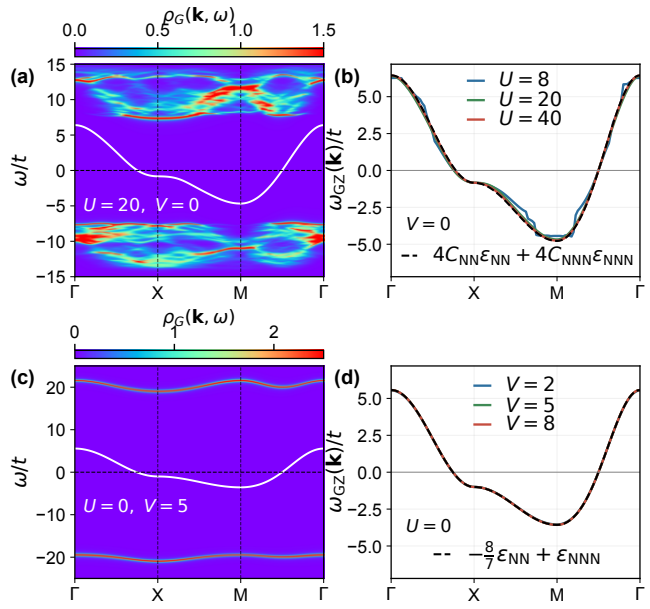


FIG. 6. **GZ dispersion at finite $t' = -0.25t$.** (a) Mott-side spectrum at $U = 20$, $V = 0$; the white curve marks the gap-interior GZ branch. (b) Corresponding U-only GZ dispersions for $U = 8, 20$, and 40, compared with the parameter-free $U \to \infty$ spin-correlation reference $4C_{\mathrm{NN}}\varepsilon_{\mathrm{NN}} + 4C_{\mathrm{NNN}}\varepsilon_{\mathrm{NNN}}$ (dashed), with $(4C_{\mathrm{NN}}, 4C_{\mathrm{NNN}}) = (-1.4029, 0.8355)$. (c) CDW-side spectrum at $U = 0$, $V = 5$. (d) Corresponding V-only GZ dispersions for $V = 2, 5$, and 8, compared with the parameter-free charge rule $-(8/7)\varepsilon_{\mathrm{NN}} + \varepsilon_{\mathrm{NNN}}$ (dashed).

has energy cost $\Delta E_{\mathrm{C}} = 7V - U$, while the particle–hole separation is $\Delta_{\mathrm{C}} = 8V - U$. Together with $\delta_{\mathbf{k}} = -\varepsilon_{\mathrm{NN}}(\mathbf{k})/\Delta E_{\mathrm{C}}$, these quantities give

$$s^{\mathrm{C}}_{\mathrm{NN}}(U,V) = -\frac{8V-U}{7V-U} + O\left(\frac{t}{\Delta E_{\mathrm{C}}}\right). \tag{9}$$

At $U = 0$, the leading term becomes the parameter-free result $-8/7$. More generally, the leading term in Eq. (9) depends only on U/V, so scaling U and V together at fixed ratio leaves the strong-coupling CDW-side coefficient unchanged. The complete derivations and numerical tests are given in Supplemental Material, Sec. 8.

The two virtual excitation energies expose the underlying physics: V softens the Mott-side doublon–holon excitation, whereas U softens the CDW-side checkerboard defect. The discontinuous change from spin-exchange physics to checkerboard-defect kinematics therefore produces the jump in s_{NN}. Independent discontinuities in the charge and spin observables and in the Hellmann–Feynman energy derivative establish the same transition as first order (Supplemental Material, Sec. 9, Figs. S8–S10, and Tables S5 and S6). Figure 5(b) shows $(V/U)_c$ decreasing from 0.2574 at $U/t = 4$ to 0.2504 at $U/t = 40$, approaching the atomic strong-coupling value of 1/4.

Frustrated hopping

Having established the NN relation at $t' = 0$, we add next-nearest-neighbor hopping to break particle–hole symmetry and determine how the GZ dispersion generalizes. Figure 6(a) shows that the resulting asymmetric Mott spectrum still contains a smooth gap-interior GZ. Writing $\varepsilon_{\mathbf{k}} = \varepsilon_{\mathrm{NN}}(\mathbf{k}) + \varepsilon_{\mathrm{NNN}}(\mathbf{k})$, the one-coefficient relation in Eq. (3) generalizes to separate contributions from the two hopping classes. The same cancellation between the addition and removal first moments applies to each class, giving

$$\omega_{\mathrm{GZ}}(\mathbf{k}) = 4\sum_b C_b\varepsilon_b(\mathbf{k}) + O(t^2_{\max}/U), \tag{10}$$

where $b = \mathrm{NN}, \mathrm{NNN}$, $C_b = \langle \mathbf{S}_i \cdot \mathbf{S}_{i+\boldsymbol{\delta}}\rangle_{\boldsymbol{\delta}\in b}$, and $t_{\max} = \max(|t|, |t'|)$. The factor 4 is unrelated to the coordination number: each of the two first-moment contributions produces $2C_b\varepsilon_b$. The full derivation is given in Supplemental Material, Sec. 5.

At large U, the corresponding spin background is the J_1–J_2 Heisenberg model, with $J_2/J_1 = (t'/t)^2$. Because $C_{\mathrm{NN}} < 0$, the NN component is inverted. Diagonal sites belong to the same Néel sublattice and have $C_{\mathrm{NNN}} > 0$, so the NNN component retains the sign of $\varepsilon_{\mathrm{NNN}}$. The numerical GZ dispersions in Fig. 6(b) stabilize with U toward this independently calculated spin-correlation prediction. Further checks for $t' = -0.25t$ and $-0.4t$ are provided in Supplemental Material, Sec. 10 and Figs. S11–S14.

The CDW-side spectrum in Fig. 6(c) likewise retains a smooth GZ between asymmetric addition and removal bands, but its two hopping contributions have a different origin. A NN hop creates the $7V$ defect that produces the $-8/7$ coefficient, whereas a NNN hop propagates an added or removed charge within one checkerboard sublattice and enters directly with unit coefficient. Consequently,

$$\omega^{(V)}_{\mathrm{GZ}}(\mathbf{k}) = -\frac{8}{7}\varepsilon_{\mathrm{NN}}(\mathbf{k}) + \varepsilon_{\mathrm{NNN}}(\mathbf{k}) + O(t^2_{\max}/V). \tag{11}$$

The V-only sequence in Fig. 6(d) approaches this parameter-free result; its derivation and finite-V corrections are given in Supplemental Material, Sec. 6. Thus, t' does not merely deform the GZ through a rescaled bare band. It separates contributions whose coefficients encode equal-time spin correlations on the Mott side and local charge-defect kinematics on the CDW side.

DISCUSSION

A GZ inside an interaction gap is not merely an absence of spectral weight: its dispersion retains an $O(t)$ record of the many-body state that forms the gap. In the

two strong-coupling limits, this record becomes quantitative. Mott-side spin correlations set the hopping-channel coefficients $s_b = 4C_b$, whereas checkerboard defect kinematics fixes $s_{\rm NN} = -8/7$ and $s_{\rm NNN} = 1$ on the CDW side. The GZ relations were obtained independently from strong-coupling expansions of the underlying extended-Hubbard Hamiltonian. In mean-field treatments, zeros, dispersive or otherwise, are absent and hence are inadequate descriptors of Mott physics. At leading strong-coupling order, increasing the mixing resolution therefore refines the numerical coefficients $s_b = 4\langle \mathbf{S}_i \cdot \mathbf{S}_j \rangle_b$ toward their thermodynamic values without changing the momentum form of $\omega_{\rm GZ}$; the CDW-side coefficients are instead fixed by local defect geometry. Agreement between the numerical Green functions and these independently derived limits establishes a new paradigm. Namely, dispersive zeros encode ordering tendencies, thereby completing the Mott paradigm. Poles tell us about quasiparticles but it is through zeros that the accompanying order of Mott physics is embedded.

At finite (U, V), the Mott-side magnitude grows as V/U approaches $(V/U)_c \gtrsim 1/4$ before switching discontinuously to the CDW-side branch. This switch coincides with discontinuities in $n(\mathbf{k})$, kinetic energy, spin correlations, and checkerboard charge correlations that identify a first-order Mott–CDW transition for large U. The dynamical zero and the underlying ground state therefore reorganize together. Although both insulators lack low-energy quasiparticle poles, their GZ dispersions distinguish the spin-correlated and checkerboard state.

This mechanism survives the breaking of particle–hole symmetry. Figure 6 shows that the NN and NNN components of the GZ separately approach the correlation-controlled values on the Mott side and the defect-kinematic values on the CDW side. Moreover, the Mott-side relation remains quantitatively down to $U = 8t$, a coupling widely used in one-band simulations of the cuprates [30, 37]. This robustness makes the GZ dispersion a practical organizing principle for momentum–frequency spectra: while poles have broadened into continua, the GZ supplies a clean, single-valued trajectory constrained by equal-time correlations. Since the complete single-particle spectrum determines $\mathrm{Re}\, G$ through a Kramers-Krönig transform, the predicted sign-change trajectory is, in principle, accessible to real-frequency many-body solvers and sufficiently complete momentum-resolved spectroscopy. Beyond cuprates, the same correlation-to-zero correspondence offers a route to expose hidden organization in other correlated insulators. In topological Mott insulators, boundary zeros have been attributed to spinon excitations[15]. Applying the present framework to boundary Green functions could directly test this attribution by determining whether the zero dispersion tracks the proposed spinon spectrum or instead obeys an equal-time correlation rule analogous to those derived here. In addition, the evolution of zeros under doping is a particularly important next problem: tracking how the GZ dispersion deforms, fragments, or reconnects with pole surfaces could reveal how a correlated insulator evolves into pseudogapped and possibly strange metallic states. In essence, the zeros are the fingerprints of ordering in strongly correlated quantum matter.

MATERIALS AND METHODS

Ground state and Green functions. For each **f**-resolved momentum-mixing sector we obtain the half-filled ground state and evaluate the single-particle Green function by Lanczos exact diagonalization. Calculations use $N_\uparrow = N_\downarrow = 8$ at the 4×4 mixing resolution. Here $Q = (0, 0)$ is the total translation quantum number of the half-filled state in the 16-mode sector; particle addition and removal enter the corresponding $Q \pm q$ sectors. The checkerboard atomic background is represented by a translation-symmetric combination of its two sublattice patterns. Each particle-addition/removal continued fraction contains at most 1000 coefficients. The spectral function is $\rho_G = -2\,\mathrm{Im}(G^h + G^p)$, and the real part used for Eq. (2) is reconstructed from the same continued fractions [38]. Frequency windows are enlarged with interaction strength so that the central gap and both adjacent spectral features are retained.

Data and code availability. Machine-readable source data for every plotted curve and the analysis code underlying the figures will be made publicly available upon publication of this paper. The original MMHK implementation and source data are openly available with Ref. [26].

Author contributions. P.M. performed the calculations and data analysis. P.M. and P.W.P. developed the interpretation and wrote the manuscript. P.W.P. supervised the project.

Competing interests. The authors declare no competing interests.

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